

Introduction to simulations workshop

Paul Whitford

10/6/22

Objective

- We are going to perform some simulations of a small protein
 - Using the C-alpha model
 - Simulate it at low temperature - This *should* stay folded.
 - Simulate it at high and low temperatures
 - This *should* unfold and refold

“should” is in italics, since there is a random element in any biological system. Something may, or may not, happen in a given simulation (it is statistical).

Software to be used

- smog2 – package that reads a PDB file and then defines a structure-based “SMOG” model
- Python – we have already installed libraries called OpenMM and OpenSMOG, which will be used to perform the simulation
- vmd – “visual molecular dynamics” allows us to visualize the trajectories

Caution!

- Commands will be shown in the font `courier`
- DO NOT COPY AND PASTE COMMANDS.
Hidden characters will interfere with the commands. Must type them!

Controlling temperature

- Temperature is the parameter that governs the average kinetic energy

$$\left\langle \frac{mv^2}{2} \right\rangle = \frac{3}{2} k_B T$$

- Left side is the average kinetic energy of a single atom.
 k_B is the Boltzmann constant
- T is the temperature
- In a simulation, specifying a temperature is accomplished by ensuring that the average kinetic energy is correct

Let's explore the role of temperature in molecular dynamics

Simulate a protein at various temperatures using a structure-based model

At low temperature, the molecule should be quite static

At high temperature, the molecule should be very mobile, and possibly unfold

Simulate a protein

Steps to get started:

- Start VirtualBox
- Launch the workshop image
- Login (if needed)
- Open a terminal

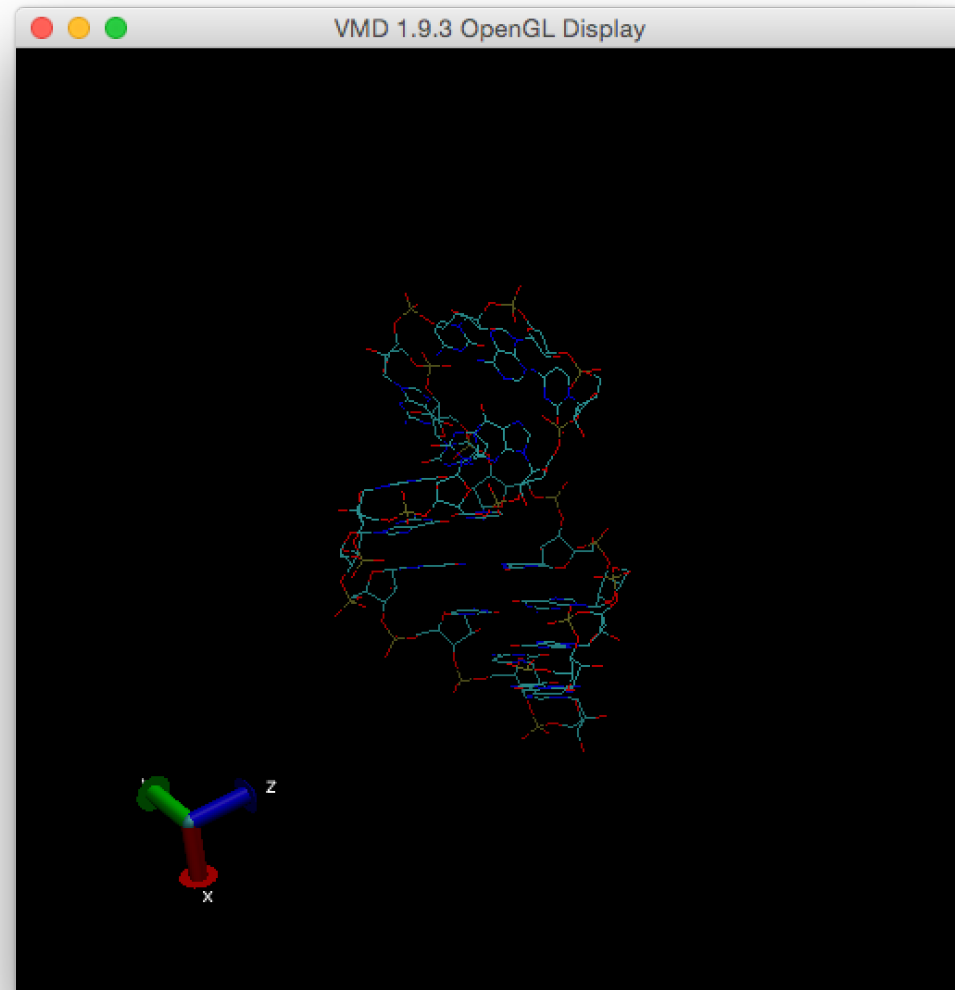
When in the terminal

- Open Firefox
- Download `https://smog-server/workshop/sims.tar`
- “untar” the file. It may be in your Download dir.
 - `tar xvf sims.tar`
- This will create a directory `sim-workshop-files`
- `cd` into `sim-workshop-files`
- When inside, use “ls” to see the contents
 - You should see some py files, and a pdb file

Simulate a protein

Visualize the protein

– `vmd 2ci2.pdb`



Simulate a protein

First, we need to generate a model. For this we will use the “SMOG” software to generate an all-atom structure-based model. However, before we can do that, we need to clean up the PDB file:

```
command: smog_adjustPDB -i 2ci2.pdb -o 2ci2.adjusted.pdb -removewater
```

This will strip the water and make sure that the atom definitions are complete.

Next, we need to generate a potential energy function.

```
command: smog2 -CA -i 2ci2.adjusted.pdb -dname 2ci2.CA -OpenSMOG
```

“-CA” indicates that we want to use the C-alpha model

“-i 2ci2.adjusted.pdb” indicates that we want to generate a model based on this pdb file

“-dname 2ci2.CA” indicates that all output files should be named 2ci2.CA.<suffix>

“-OpenSMOG” indicates that we want to generate files for a specific simulation tool (OpenSMOG)

If smog is successful, you can use “ls” and see several new files

gro (coordinate) file

Gro file for a structure based model, generated with SMOG Version 2.0.4t

1230

1C	P	1	-0.449	-1.744	-1.435
1C	O1P	2	-0.366	-1.748	-1.558
1C	O2P	3	-0.427	-1.637	-1.335
1C	O5*	4	-0.604	-1.742	-1.478
1C	C5*	5	-0.642	-1.752	-1.616
1C	C4*	6	-0.793	-1.765	-1.635
1C	O4*	7	-0.852	-1.849	-1.537
1C	C3*	8	-0.864	-1.632	-1.619
1C	O3*	9	-0.876	-1.573	-1.749
1C	C2*	10	-1.005	-1.671	-1.571
1C	O2*	11	-1.100	-1.664	-1.678
1C	C1*	12	-0.989	-1.812	-1.522
1C	N1	13	-1.032	-1.821	-1.382
1C	C2	14	-1.167	-1.827	-1.359
1C	O2	15	-1.245	-1.824	-1.454
1C	N3	16	-1.211	-1.838	-1.230
1C	C4	17	-1.123	-1.842	-1.129
1C	N4	18	-1.168	-1.853	-1.004
1C	C5	19	-0.982	-1.835	-1.153
1C	C6	20	-0.942	-1.825	-1.281
2U	P	21	-0.944	-1.428	-1.769
2U	O1P	22	-0.915	-1.382	-1.907
2U	O2P	23	-0.906	-1.343	-1.654
2U	O5*	24	-1.102	-1.460	-1.758
2U	C5*	25	-1.178	-1.494	-1.875
2U	C4*	26	-1.327	-1.471	-1.852

.top (forcefield) file

Indicates which atom types are in this simulation

Name of the complex

Specific atoms in this system

List of native contacts, native distances and energetic weights

Chemical bonds

internal bookkeeping

bond angles

dihedrals

```
[ defaults ]
;nbfunc comb-rule gen-pairs
      1          1 no

[ atomtypes ]
;name mass      charge      ptype c6      c12
CA    1.000      0.000 A      0.000 0.16777E-04

[ moleculetype ]
;name nrexcl
Macromolecule      3

[ atoms ]
;nr  type  resnr residue atom  cgnr charge  mass
  1  CA      1  VAL  CA      1  0.0   1.000
  2  CA      2  LEU  CA      2  0.0   1.000
...
...

[ pairs ]
  1  578 1  0.30873E-02  0.56583E-03
  1  579 1  0.87833E-02  0.19842E-02
...
...

[ bonds ]
;ai  aj  func  r0(nm)  Kb
  1  2  1  0.38091E+00  0.20000E+05
  2  3  1  0.38222E+00  0.20000E+05
  3  4  1  0.38511E+00  0.20000E+05
...
...

[ exclusions ]
;ai  aj
  1  578
  1  579
  1  582
...
...

[ angles ]
;ai  aj  ak  func  th0(deg)  Ka
  1  2  3  1  0.12475E+03  0.40000E+02
  2  3  4  1  0.11047E+03  0.40000E+02
  3  4  5  1  0.10419E+03  0.40000E+02
...
...

[ dihedrals ]
;ai  aj  ak  al  func  phi0(deg)  Kd(kJ/mol)  mult
  1  2  3  4  1  0.44326E+03  0.10000E+01  1
  1  2  3  4  1  0.13298E+04  0.50000E+00  3
  2  3  4  5  1  0.43680E+03  0.10000E+01  1
...
...

[ system ]
;name
Macromolecule

[ molecules ]
Macromolecule 1
```

Simulate the protein at low temperature

- We will use OpenSMOG/OpenMM, which are python libraries. Since we have already prepared the scripts, we can run the simulation with a simple command:
 - `python run.CA.py`
- Before we do, let look at run.CA.py. Open the file with:
 - `gedit run.CA.py`
- After this, let's actually run the system

Visualize your simulation

- Launch VMD
- Load 2ci2.CA.gro
- Show in tube representation
- Load the trajectory file:
`output_2ci2.CA/2ci2_trajectory.dcd`
- What does the simulation look like? Does it unfold?

Simulate the protein at high and then low temperature

- We will use OpenSMOG/OpenMM, which are python libraries. Since we have already prepared the scripts, we can run the simulation with a simple command:
 - `python run.CA.high-low.py`
- Before we do, let look at the py file. Open the file with:
 - `gedit run.CA.high-low.py`
- After this, let's actually run the system

Visualize your simulation

- Launch VMD
- Load 2ci2.CA.gro
- Show in tube representation
- Load the first part of the trajectory file:
`output_2ci2.CA.hot/2ci2_trajectory.dcd`
- What does the simulation look like? Does it unfold?
- Now also load the second part of the trajectory:
`output_2ci2.CA.cold/2ci2_trajectory.dcd`
- What does the simulation look like? Does it fold?

Congrats!

- You have just simulated protein folding and unfolding!